Juniper Core Node Failure SOP for Hardware/Environmental Faults on BLR-CORE-01

Node: BLR-CORE-01 | **Vendor:** Juniper | **Severity:** Critical | **Alarm Code:** ALARM_3247

Description

This Standard Operating Procedure (SOP) aims to troubleshoot and resolve critical hardware/environmental faults on the Juniper core node BLR-CORE-01. The core node is a critical component of the Telecom network, responsible for high-speed routing and switching. Hardware/environmental faults in this role can cause significant network disruptions and data loss. The fault in question is caused by ASIC memory parity errors, line card crashes, or power supply redundancy loss. The Juniper Ex Series routers employ ASIC-based hardware, which is prone to memory parity errors due to the complex nature of high-speed switching and routing. The line cards in these routers are also susceptible to crashes due to overheating, electrical surges, or faulty components. Additionally, the power supply redundancy loss can occur due to failure of one of the redundant power supplies or improper configuration. The Juniper JUNOS operating system provides a robust CLI for troubleshooting and resolving these faults. This SOP outlines the necessary steps to diagnose and resolve the fault, ensuring minimal network downtime and data loss.

Pre-requisite Checks

- Check: Verify the site location and ensure remote hands are dispatched to the Unknown Site., Description: Before initiating any troubleshooting steps, it is essential to confirm the site location and ensure that remote hands are dispatched to the site to assist with any physical checks or repairs.

- Check: Verify the redundant link status to ensure network redundancy., Description: To minimize network downtime, it is crucial to verify the status of redundant links to ensure that they are functioning correctly and can take over in case of a primary link failure.

Potential Root Causes

Cause	Probability
ASIC memory parity errors	High
Line card crashes due to overheating or electrical surges	Medium
Power supply redundancy loss due to faulty components or improper configuration	Low

Diagnostic Commands & Expected Outputs

```
Command: show chassis environment, Expected Output: Temperature sensor values, power supply redundancy status, and other environmental parameters, Description: This command displays the environmental parameters of the router, including temperature sensor values, power supply redundancy status, and other relevant data. This information helps identify potential causes of the fault.
```

```
Command: show chassis fpc statistics, Expected Output: FPC statistics, including error counts and alarm status, Description: This command displays the FPC statistics, including error counts and alarm status. This information helps identify potential causes of the fault, such as ASIC memory parity errors or line card crashes.

Command: show system alarm, Expected Output: System alarm messages, including alarm codes and descriptions, Description: This command displays system alarm messages, including alarm codes and descriptions. This information helps identify the specific alarm code (ALARM_3247) and its corresponding description, which can aid in troubleshooting.
```

Resolution Steps

Step 1: Step: Power cycle the router, Command: request system reboot, Description: Power cycling the router can resolve issues caused by software or hardware malfunctions. This step should be performed with caution, as it may cause temporary network downtime.

Step 2: Step: Replace the faulty FPC, Command: request chassis replace fpc, Description: Replacing the faulty FPC can resolve issues caused by ASIC memory parity errors or line card crashes. This step requires physical access to the router and should be performed by a qualified technician.

Step 3: Step: Replace the faulty power supply, Command: request chassis replace power-supply, Description: Replacing the faulty power supply can resolve issues caused by power supply redundancy loss. This step requires physical access to the router and should be performed by a qualified technician.

Post-Check Verification

```
Command: show chassis environment, Expected Output: Temperature sensor values, power supply redundancy status, and other environmental parameters, Description: This command verifies that the environmental parameters are within normal ranges and the power supply redundancy status is stable.

Command: show chassis fpc statistics, Expected Output: FPC statistics, including error counts and alarm status, Description: This command verifies that the FPC statistics are normal and the alarm status is clear.

Command: show system alarm, Expected Output: System alarm messages, including alarm codes and descriptions, Description: This command verifies that the system alarm messages are clear and there are no active alarms.
```

Roll-back Procedures

- Step: Restore the original configuration, Command: load configuration /config/ Juniper Networks router-name.conf, Description: This step restores the original configuration of the router, which can help resolve issues caused by incorrect configuration changes.

- Step: Reset the router to factory defaults, Command: request system factory-default, Description: This step resets the router to its factory defaults, which can help resolve issues caused by software or hardware malfunctions.